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COMPUTER SCIENCE DEPARTMENT

FINAL PROJECT PROPOSAL

Medical **AI** Prediction System
(**MAPS**)

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1. Introduction

The Medical AI Prediction System (MAPS) is a comprehensive, role-based medical platform designed to integrate artificial intelligence into healthcare delivery while maintaining strict professional governance and clinical oversight. This proposal presents the design, architecture, and implementation plan for MAPS, a system that combines ML.NET-based disease prediction, ONNX-powered medical image classification, NLP-driven clinical note processing, and real-time SignalR communication within a unified platform governed by a hierarchical access control model.

MAPS addresses a critical gap in healthcare technology: the absence of unified platforms that combine AI-driven diagnostics with role-appropriate access controls. Existing solutions typically offer either AI prediction capabilities or hospital management features, but rarely both within a single integrated ecosystem. Our platform bridges this divide by providing a layered architecture where Administrators hold the highest authority for system governance, Doctors serve as the clinical backbone with advanced AI-assisted tools, and Patients interact through a controlled, secure portal. Critically, the Patient Module does not include direct AI chatbot access; all AI-generated insights are routed through assigned doctors to maintain clinical accountability and prevent misinterpretation of diagnostic results.

1.1 Project Objectives

The primary objectives of the MAPS project are to develop a functional medical AI platform that demonstrates the integration of multiple AI paradigms within a clinically responsible framework. The following table outlines the core project objectives and their associated success criteria.

Objective	Description	Success Criteria
AI-Driven Diagnostics	Provide multi-condition differential diagnosis with probability rankings and recommended confirmatory tests	Prediction accuracy exceeds 85% on validation data
Proactive Risk Management	Continuously assess patient risk scores and generate alerts before conditions deteriorate	Alert system responds within 5 minutes of threshold breach
Administrative Oversight	Enable comprehensive user management, audit trails, and system analytics for administrators	Complete audit trail for all system actions
Secure Communication	Facilitate role-hierarchical real-time chat with permission-based access enforcement	Zero unauthorized access incidents
Clinical NLP	Extract and structure medical entities from clinical notes automatically using NLP	Entity extraction F1 score exceeds 0.80
Medical Image Analysis	Analyze X-rays and medical scans using CNN models to highlight potential abnormalities	Image classification accuracy exceeds 80%

Smart Scheduling	Prioritize appointments based on AI risk scores for optimal resource allocation	High-risk patients receive same-day appointments
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Table 1: MAPS Project Objectives and Success Criteria

2. Role-Based Hierarchy System

The MAPS platform implements a strict three-tier role hierarchy that governs every aspect of system access, data visibility, and feature availability. This hierarchical model is a foundational architectural principle ensuring clinical accountability, data security, and operational integrity. Administrators possess unrestricted system oversight and control, Doctors serve as clinical decision-makers with advanced AI tools, and Patients interact through a controlled portal providing health information without direct AI diagnostic access.

The decision to remove the AI chatbot from the Patient role is clinically motivated. Allowing patients to interact directly with AI diagnostic tools without professional mediation introduces medical liability risks, potential for misinterpretation of results, and the danger of self-diagnosis leading to harmful health decisions. By routing all AI-generated insights through assigned doctors, MAPS ensures that every prediction, risk assessment, and diagnostic suggestion is contextualized and validated by a qualified medical professional before reaching the patient.

2.1 Role Hierarchy Overview

The following diagram illustrates the three-tier access control model, showing the hierarchical relationship between roles and their respective permissions within the MAPS platform. Each tier inherits restrictions from the tier above, ensuring that data flows downward through proper clinical mediation channels.

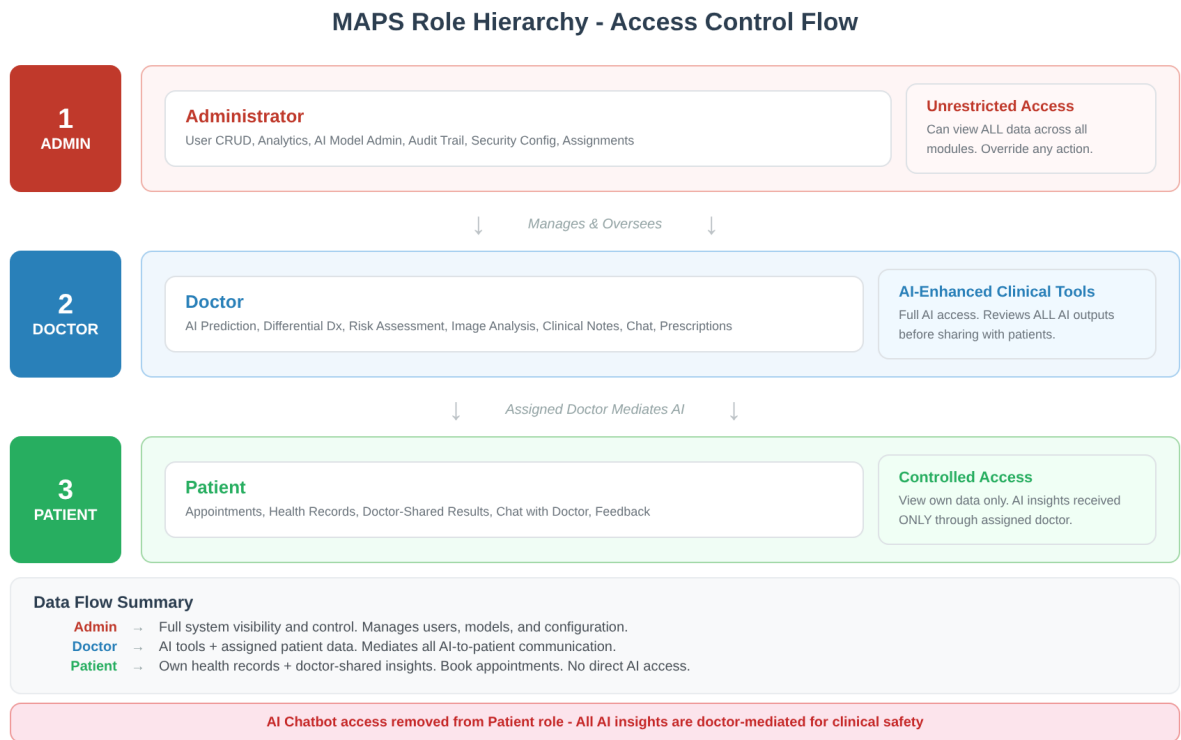


Figure 1: MAPS Role Hierarchy - Three-Tier Access Control Flow

2.2 Role Feature Comparison

The following matrix provides a comprehensive comparison of feature access across all three roles. The AI Chatbot row is highlighted to indicate that this feature has been removed from all roles, with AI-driven insights provided exclusively through the Doctor Module.

Feature / Capability	Admin (Tier 1)	Doctor (Tier 2)	Patient (Tier 3)
User Management (CRUD)	Full Access	View Own Profile	View Own Profile
Doctor-Patient Assignment	Full Control	View Assignments	View Assignment
System Analytics Dashboard	Full Access	Department Stats	No Access
AI Model Administration	Full Control	Use Models	No Access
Audit Trail and Compliance	Full Access	View Own Logs	No Access
System Configuration	Full Control	No Access	No Access
AI Disease Prediction	View All Results	Full Access	Doctor-Shared Only
AI Differential Diagnosis	View All Results	Full Access	Doctor-Shared Only
Patient Risk Assessment	View All Alerts	Full Access	Doctor-Shared Only
Clinical Notes + AI Summary	View All Notes	Full Access	View Own Notes

Medical Image Analysis	View All Results	Full Access	Doctor-Shared Only
Smart Appointment Priority	Override Priority	Full Access	Book (AI-Priority)
Doctor-Patient Chat	Chat with Anyone	Chat with Assigned	Chat with Assigned
AI Chatbot	No Access	No Access	No Access
Prescription Management	View All	Create and Manage	View Own
Feedback System	View All Feedback	View Own Feedback	Submit Feedback
Health Records	View All Records	View Assigned Records	View Own Records
Announcements	Create and Manage	View Announcements	View Announcements

Table 2: Role-Based Feature Comparison Matrix (AI Chatbot row highlighted as removed)

2.3 Chat Permission Matrix

Initiator	Can Chat With Admin	Can Chat With Doctor	Can Chat With Patient
Admin	Yes (All Admins)	Yes (Any Doctor)	Yes (Any Patient)
Doctor	Yes (Any Admin)	Yes (Assigned Colleagues)	Yes (Assigned Patients Only)
Patient	Yes (Support Channel)	Yes (Assigned Doctor Only)	No

Table 3: Chat Permission Matrix by Role

3. System Architecture

The MAPS platform follows a layered architecture pattern that separates concerns across presentation, business logic, AI/ML processing, and data persistence layers. This architecture ensures modularity, scalability, and maintainability while supporting the complex interactions between AI models, real-time communication, and role-based access control. The system is built on the .NET ecosystem with ASP.NET Core for the web application, ML.NET for disease prediction, ONNX Runtime for image classification, and SignalR for real-time bidirectional communication.

3.1 Architecture Diagram

The following diagram presents the four-layer architecture of the MAPS platform, showing the relationships between the Presentation, Application, AI/ML, and Data layers, along with the cross-cutting concerns that span all layers.

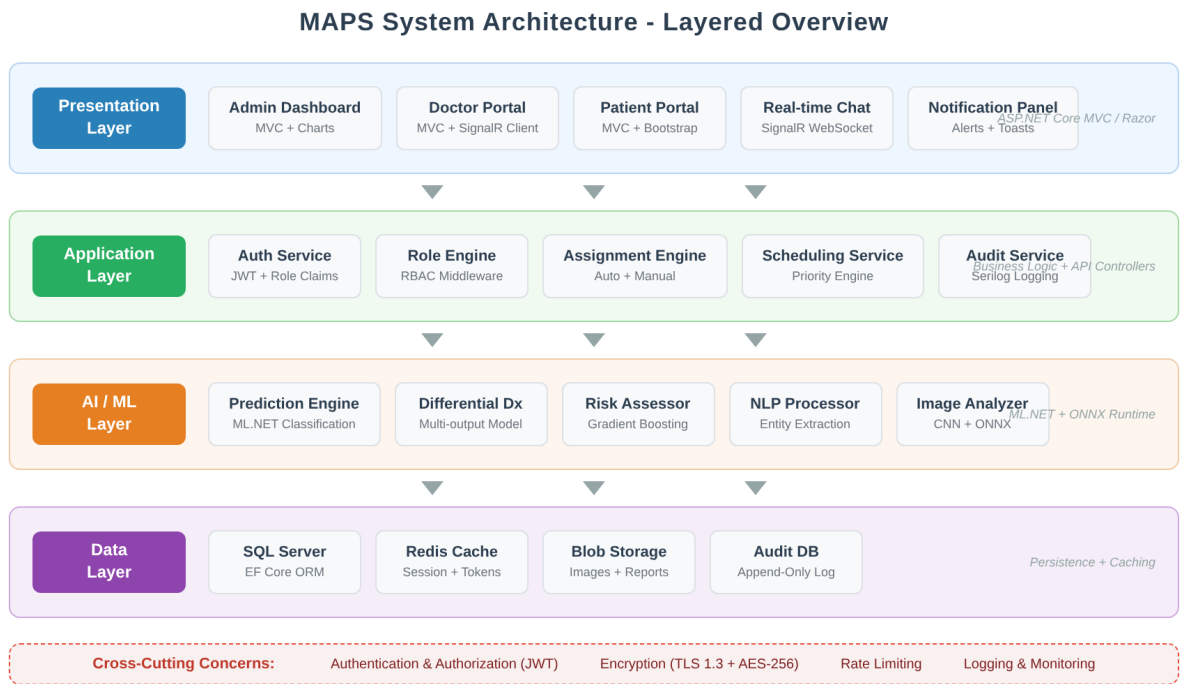


Figure 2: MAPS System Architecture - Layered Overview

3.2 Technology Stack

Layer	Technology	Purpose
Frontend	ASP.NET Core MVC / Razor Pages	Server-side rendered UI with responsive design
Styling	Bootstrap 5 + Custom CSS	Responsive layout and component library
Charts	Chart.js / ApexCharts	Interactive data visualization dashboards
Backend	ASP.NET Core Web API	RESTful API endpoints and business logic
Real-time	SignalR	Bidirectional WebSocket communication for chat and alerts
AI Prediction	ML.NET + ONNX Runtime	Disease prediction and image classification models
NLP	ML.NET + Custom Pipeline	Clinical note summarization and entity extraction
Authentication	ASP.NET Core Identity + JWT	Secure authentication with role-based claims
Database	SQL Server / Entity Framework Core	Relational data persistence with ORM

Caching	Redis / In-Memory Cache	Session management and performance optimization
File Storage	Azure Blob / Local Storage	Medical image and document storage
Logging	Serilog + Seq	Structured logging with audit trail

Table 4: MAPS Technology Stack

4. Module A: Admin Panel

The Admin Panel is the command center of the MAPS platform, providing administrators with comprehensive oversight and control over every aspect of the system. As the highest tier in the role hierarchy, administrators have unrestricted access to user management, system analytics, AI model configuration, audit trails, and security settings. The Admin Panel is organized into eight distinct sub-sections, each addressing a critical administrative function.

4.1 User Lifecycle Management

The User Lifecycle Management module provides administrators with complete control over user accounts from creation through deactivation. This includes the ability to create new doctor and patient accounts, assign roles and permissions, manage account activation and deactivation, and handle password resets. The module supports bulk operations for efficient management of large user bases, including CSV import for batch user creation and bulk status updates. Every user lifecycle event is logged in the audit trail, ensuring complete accountability for administrative actions. The registration process incorporates a Registration Lock mechanism that requires admin approval before new accounts become active, preventing unauthorized self-registration and ensuring that only verified medical professionals and validated patients gain system access.

4.2 Doctor-Patient Assignment Engine

The Doctor-Patient Assignment Engine manages the critical relationship between doctors and their assigned patients. Administrators can create, modify, and terminate assignments, transfer patients between doctors, and view assignment histories. The engine supports both manual assignment through an administrative interface and rule-based auto-assignment that considers factors such as doctor specialization, current patient load, and department availability. When a patient is transferred between doctors, all associated medical records, prescriptions, and AI prediction results are automatically linked to the new assignment, ensuring continuity of care. The assignment engine also enforces business rules such as maximum patient load per doctor and mandatory specialization matching for certain condition types.

4.3 System Analytics Dashboard

The System Analytics Dashboard provides administrators with a real-time overview of platform performance, user activity, and clinical metrics. The dashboard features interactive charts displaying key performance indicators such as total user counts by role, active sessions, prediction volumes, and system health metrics. Administrators can filter analytics by date range, department, and individual users. The dashboard includes pre-built analytics views for User Activity Trends, AI Prediction Analytics, Appointment Statistics, and System Performance Metrics, each supporting data export in CSV and PDF formats for reporting and compliance purposes.

4.4 AI Model Administration

The AI Model Administration module gives administrators control over the machine learning models powering the platform. This includes viewing currently deployed models, monitoring performance metrics, updating model versions, and configuring prediction thresholds. Administrators can upload new ONNX model files, set confidence thresholds for different prediction types, and manage feature importance weights. Model versioning allows rollback to previous versions if a newly deployed model exhibits degraded performance. The module supports A/B testing configurations where a percentage of prediction requests can be routed to a candidate model for performance comparison before full deployment.

4.5 Feedback Analytics

The Feedback Analytics module aggregates and analyzes patient feedback submitted through the Patient Module. Administrators can view overall satisfaction scores, sentiment analysis results, and detailed feedback comments. The module provides trend analysis showing how satisfaction metrics change over time, correlation analysis between feedback scores and specific doctors or departments, and automated alerting when feedback patterns indicate potential quality of care issues. Feedback data is processed using NLP sentiment analysis to automatically categorize comments as positive, negative, or neutral, with key theme extraction identifying recurring issues.

4.6 Audit Trail and Compliance

The Audit Trail and Compliance module provides a comprehensive, tamper-proof record of every action performed within the MAPS platform. Every user login, data access event, prediction request, prescription creation, administrative action, and configuration change is logged with full context including user identity, timestamp, IP address, action type, and affected entities. The audit trail is append-only, meaning records cannot be modified or deleted even by administrators, ensuring the integrity of the compliance trail. Automated alerts are triggered for suspicious activities such as unusual data access patterns or multiple failed login attempts.

4.7 System Configuration and Security

The System Configuration and Security module provides centralized control over platform settings, security policies, and operational parameters. This includes managing registration lock settings, configuring JWT token expiration and refresh policies, setting password complexity requirements, and defining IP-based access restrictions. Security features include configurable rate limiting for API endpoints, session management policies controlling concurrent logins and idle timeout durations, IP whitelist/blacklist management, and two-factor authentication configuration for administrator accounts. All configuration changes are logged in the audit trail and require confirmation before taking effect.

4.8 Admin Dashboard Layout

The Admin Dashboard is designed as a single-page application with a persistent sidebar navigation and a dynamic content area. The sidebar provides quick access to all eight administrative sub-sections, with visual indicators for pending actions such as registration approvals, unread alerts, and compliance review items. The main content area features a customizable widget-based layout where administrators can arrange analytics cards, quick-action buttons, and notification feeds. The dashboard is fully responsive, supporting desktop, tablet, and mobile form factors.

5. Module B: Doctor Module

The Doctor Module is the most feature-rich and technically sophisticated component of the MAPS platform, comprising eleven distinct features spanning AI-powered diagnostics, real-time communication, clinical documentation, and intelligent scheduling. Five AI-driven features transform this module into a comprehensive AI-augmented clinical decision support system. Each feature interconnects with the others, creating a unified clinical workflow where AI insights flow naturally from symptom analysis through risk assessment, appointment prioritization, imaging, and documentation.

5.1 AI-Powered Clinical Workflow

The following diagram illustrates the end-to-end clinical workflow within the Doctor Module, showing how the five AI features interconnect to form a continuous loop from patient consultation through AI analysis, clinical review, and patient feedback. This workflow ensures that every AI-generated insight passes through the doctor's review before reaching the patient.

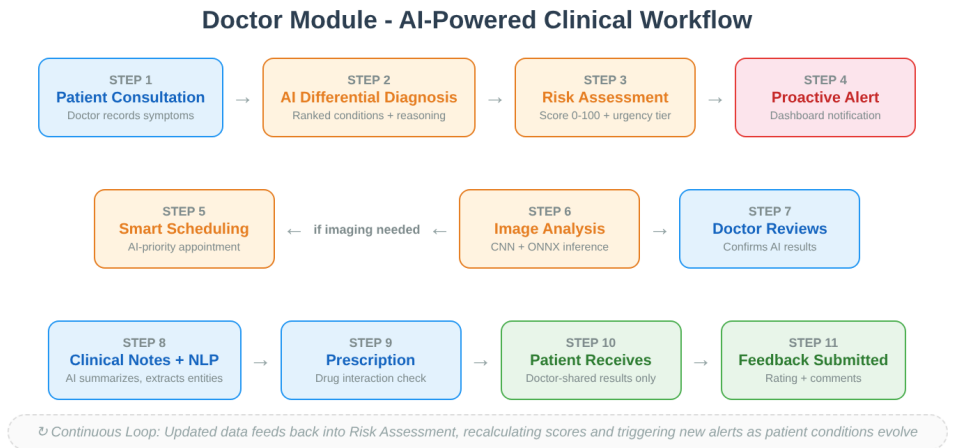


Figure 3: Doctor Module - AI-Powered Clinical Workflow

5.2 Doctor Dashboard

The Doctor Dashboard serves as the central hub for clinical operations, providing an at-a-glance overview of the doctor's daily schedule, patient statistics, pending alerts, and recent AI prediction results. The dashboard is organized into a grid layout featuring patient count cards showing total, active, and high-risk patients, an appointment timeline with priority indicators, a risk alert panel highlighting patients whose risk scores have recently increased, and a quick-access panel for recent AI predictions and pending clinical notes. The dashboard updates in real-time through SignalR connections, ensuring that doctors always see the most current information without manual page refreshes.

5.3 AI Disease Prediction Engine

The AI Disease Prediction Engine is the foundational AI feature, enabling doctors to input a patient's symptoms and receive an AI-generated prediction of the most likely condition along with a confidence percentage. The engine uses ML.NET models trained on clinical datasets to map symptom combinations to disease probabilities. Doctors interact with the prediction engine through a symptom selection interface where they can search for and select symptoms from a standardized medical vocabulary, specify symptom duration and severity, and add relevant patient history context. The engine processes these inputs through the ML.NET prediction pipeline and returns results within seconds, displayed through an intuitive interface that clearly separates the AI suggestion from the doctor's own clinical judgment.

5.4 AI-Assisted Differential Diagnosis

The AI-Assisted Differential Diagnosis feature extends the basic prediction engine by generating a ranked list of possible diagnoses rather than a single prediction. When a doctor inputs a patient's

symptoms, the system returns multiple potential conditions ranked by probability, along with reasoning paths that explain why each condition is being suggested, and recommended confirmatory or rule-out tests for each candidate. This mirrors the clinical practice of differential diagnosis, where physicians consider multiple possible explanations before arriving at a final diagnosis. The technical implementation uses a multi-output classification model that generates probability distributions across the disease catalog, then filters and ranks the top candidates based on clinical relevance thresholds.

5.5 Patient Risk Assessment and Proactive Alerts

The Patient Risk Assessment module continuously analyzes patient health data and AI prediction history to calculate dynamic risk scores ranging from 0 to 100. The risk calculation considers multiple factors including recent AI prediction results, vital sign trends, medication adherence patterns, and the severity and progression of documented symptoms. The alerting system operates on a four-tier urgency classification: Emergency (risk above 80) triggers same-day alerts, Urgent (60-80) generates within-24-hour alerts, Normal (30-60) produces within-3-day alerts, and Follow-up (below 30) creates within-1-week reminders. Each alert includes the risk score, contributing factors, trend direction, and suggested actions. This transforms the care model from reactive to predictive, identifying at-risk patients before adverse outcomes occur.

5.6 Clinical Notes with AI Summarization

The Clinical Notes module allows doctors to document consultations using free-text notes, which are then processed by an NLP pipeline that automatically summarizes the content, extracts key medical entities, and structures the information into the patient's health record. The NLP engine identifies and categorizes entities including symptoms with onset and duration, vital signs, medications with dosage, procedures, tests ordered, and follow-up instructions. The summarization engine generates a concise three-to-five sentence summary capturing key findings, decisions, and action items. Extracted entities are linked to the patient's structured health record, making the data available for AI prediction, risk assessment, and reporting. The NLP pipeline achieves entity extraction F1 scores above 0.80 on clinical documentation.

5.7 Medical Image Analysis

The Medical Image Analysis feature enables doctors to upload medical images (X-rays, CT scans, skin lesions) for AI-powered analysis using CNN models served through ONNX Runtime. The pipeline includes image preprocessing (resizing, normalization, contrast enhancement using CLAHE), CNN inference for feature extraction and classification, post-processing with non-maximum suppression, and report generation with bounding boxes and confidence scores. The doctor reviews the AI analysis, confirms or modifies findings, and adds clinical context before saving to the patient record. Supported imaging modalities include chest X-rays for pneumonia and cardiomegaly detection, skin lesion images for melanoma screening, and CT scan slices for tumor identification.

5.8 Smart Appointment Priority Engine

The Smart Appointment Priority Engine uses AI risk scores from the Risk Assessment module to intelligently prioritize appointment scheduling. Instead of first-come-first-served, the engine assigns slots based on clinical urgency: Emergency patients (risk above 80) are scheduled same-day, Urgent (60-80) within 24 hours, Normal (30-60) within three business days, and Follow-up (below 30) within one week. The engine performs conflict detection to prevent double-booking, manages doctor availability windows, and considers appointment type duration. When a patient books an appointment, the system automatically calculates their risk score and assigns the appropriate priority tier. Administrators can override priority levels when clinical circumstances warrant deviation from AI-recommended scheduling.

5.9 Patient Management

The Patient Management interface provides doctors with a comprehensive view of all their assigned patients, including demographic information, medical history, current prescriptions, recent AI predictions, and risk assessment scores. The patient list supports filtering by risk level, condition type, and appointment status. Each patient's detail view includes a timeline of all clinical interactions, AI prediction results, medication changes, and appointment history, providing a complete longitudinal health record.

5.10 Prescription Management

The Prescription Management module allows doctors to create, modify, and manage prescriptions for assigned patients. The interface supports a structured prescription form including medication selection from a drug database, dosage and frequency specification, duration of treatment, and special instructions. The module includes basic drug interaction checking that flags potential conflicts between newly prescribed medications and existing prescriptions, and provides warnings for known allergies documented in the patient's record.

5.11 Doctor-Patient Chat System

The Doctor-Patient Chat System provides real-time communication through SignalR WebSocket connections. The interface supports text messages, image sharing for wound photos or test results, and file attachments. The chat system enforces the role-based permission matrix, ensuring that doctors can only chat with assigned patients and that patients can only communicate with their assigned doctor. Chat history is persisted in the database and linked to the patient's record. Doctors can mark specific messages as clinically relevant, which automatically links them to the patient's health record timeline.

5.12 Medical Reports Generation

The Medical Reports module enables doctors to generate comprehensive clinical reports that aggregate patient health data, AI prediction results, risk assessment outcomes, and treatment plans into formatted

documents. Reports can be generated for individual patients or as aggregate statistical reports. The module supports multiple report formats including detailed clinical summaries, referral letters, and statistical dashboards. All generated reports can be exported as PDF documents.

6. Module C: Patient Module

The Patient Module provides patients with a secure, intuitive interface for managing their healthcare interactions. Designed with controlled access, the module ensures that patients can view personal health information, communicate with their assigned doctor, and manage appointments, while all AI-generated insights are routed through their doctor to maintain clinical oversight. The Patient Module does not include an AI chatbot feature; all AI predictions, risk assessments, and diagnostic suggestions are shared by the doctor during consultations or through the doctor-mediated results viewing system. This design ensures that AI serves as a clinical support tool rather than a self-service diagnostic mechanism.

6.1 Patient Dashboard

The Patient Dashboard provides a welcoming, easy-to-navigate interface displaying upcoming appointments, recent health updates, and messages from the assigned doctor. The dashboard features appointment cards with date, time, and priority indicators, a health summary panel showing key vitals and active prescriptions, a messaging widget for quick communication, and a notification feed for system announcements and doctor-shared results. The interface is designed with accessibility in mind, featuring large touch targets, high-contrast color schemes, and clear typography.

6.2 Appointment Booking

Patients can book, reschedule, and cancel appointments through an intuitive scheduling interface. When a patient initiates a booking, the Smart Appointment Priority Engine automatically calculates their priority tier based on their current risk score and presents available slots accordingly. High-risk patients are offered earlier time slots, while routine follow-up patients are shown standard availability. The booking system prevents double-booking and manages doctor availability in real-time through SignalR synchronization.

6.3 Health Records and History

The Health Records module provides patients with read-only access to their complete medical history within the MAPS platform. This includes consultation notes as shared by the doctor, prescription records, AI prediction results that the doctor has chosen to share, lab test results, and imaging reports. Records are organized chronologically in a timeline format. Patients can download their records in PDF format for personal archiving or sharing with external healthcare providers.

6.4 View AI Predictions (Doctor-Shared Only)

Patients can view AI prediction results only when their assigned doctor has explicitly shared them. When a doctor shares a prediction, they add clinical context and interpretation that helps the patient understand what the AI analysis means. The shared prediction view includes the primary condition identified, the confidence percentage in a patient-friendly format, the doctor's interpretation and recommended next steps, and any lifestyle modifications or follow-up tests recommended. This mediated sharing approach ensures that patients receive AI insights within the appropriate clinical context.

6.5 Feedback System

The Feedback System allows patients to rate their consultation experience and provide detailed feedback on the care they received. The feedback form includes a star rating scale, structured evaluation categories (communication quality, wait time, overall experience), and a free-text comment field. Feedback submissions are anonymous to the assigned doctor by default. All feedback data is aggregated and analyzed through the Admin Module's Feedback Analytics dashboard.

7. AI/ML Pipeline

The MAPS AI/ML pipeline encompasses four distinct processing streams that serve different clinical functions: the Disease Prediction Pipeline, the Differential Diagnosis Pipeline, the Risk Assessment Pipeline, and the Medical Image Analysis Pipeline. Each pipeline is independently deployable and scalable, allowing the system to allocate computational resources based on demand patterns without affecting other AI features.

7.1 Pipeline Data Flow

The following diagram illustrates the data flow through all four AI/ML pipelines, showing how each pipeline processes its inputs, performs model inference, and produces outputs that feed into the clinical workflow. The feedback loop at the bottom indicates how prediction results and risk scores feed back into the data pool, improving future model accuracy through periodic retraining.

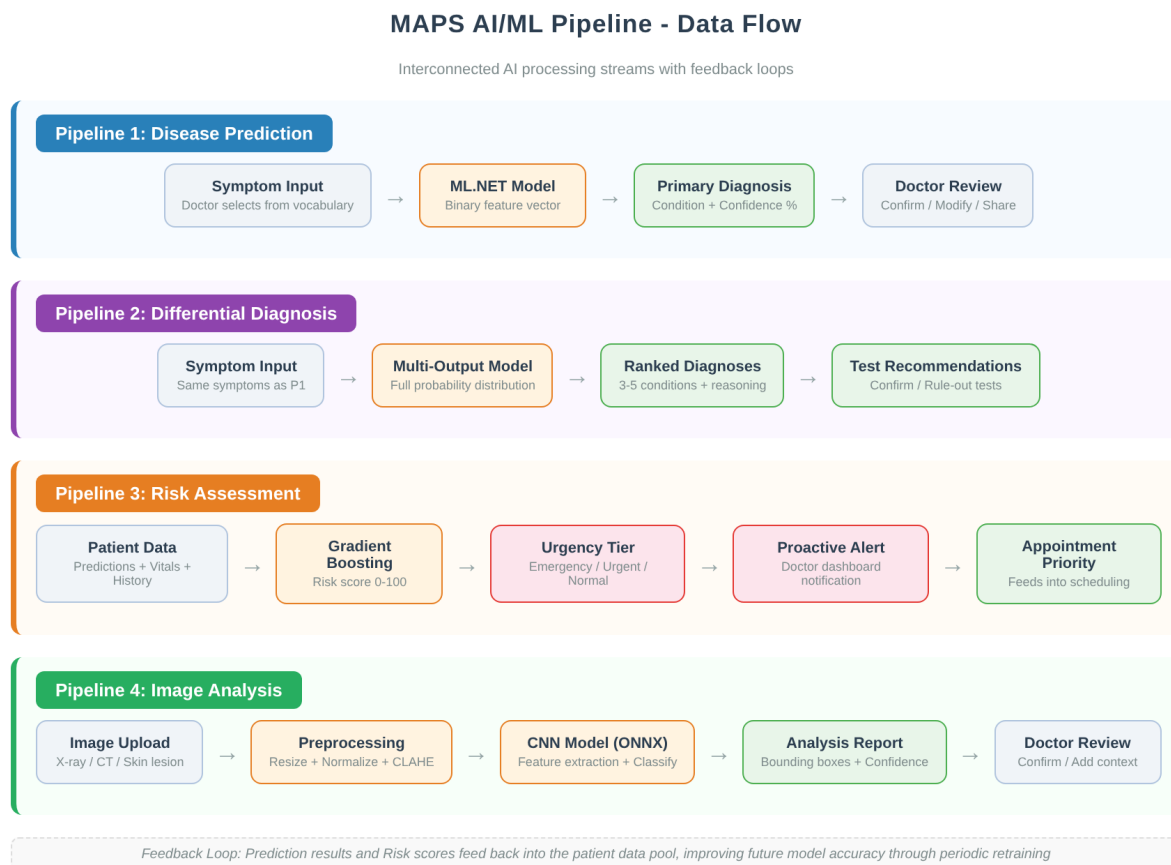


Figure 4: MAPS AI/ML Pipeline - Data Flow

7.2 Disease Prediction Pipeline

The Disease Prediction Pipeline processes symptom inputs through a multi-stage workflow: symptom encoding converts the selected symptoms into a binary feature vector, the ML.NET prediction model applies the trained classification algorithm to generate probability distributions across the disease catalog, and the result formatting stage selects the highest-probability diagnosis, extracts the top contributing features, and formats the output for display. The model is trained on curated clinical datasets with feature engineering that includes symptom duration weighting, co-occurrence frequency analysis, and demographic adjustment factors.

7.3 Differential Diagnosis Pipeline

The Differential Diagnosis Pipeline extends the base prediction pipeline by retaining the full probability distribution rather than selecting only the top result. Clinical filtering rules remove statistically possible but clinically implausible diagnoses based on patient demographics and symptom patterns, the remaining candidates are ranked by a composite score balancing model probability with clinical relevance, and reasoning chains are generated for each candidate by extracting attention weights that link specific symptoms to the diagnosis. Each differential diagnosis entry includes recommended

confirmatory tests mapped from clinical practice guidelines.

7.4 Risk Assessment Pipeline

The Risk Assessment Pipeline operates as a background process that recalculates patient risk scores whenever new clinical data enters the system. The pipeline aggregates features from multiple sources including recent AI prediction outcomes, vital sign trends from clinical notes, appointment attendance patterns, prescription adherence indicators, and the patient's overall health trajectory. A gradient boosting model processes these features to produce a risk score between 0 and 100, which is then classified into the four-tier urgency system and compared against the previous score to determine trend direction.

7.5 Medical Image Analysis Pipeline

The Medical Image Analysis Pipeline handles uploaded medical images through a computer vision workflow: image preprocessing (resizing to model input dimensions, pixel normalization, and contrast enhancement using CLAHE), inference through a CNN model that outputs classification probabilities and bounding box coordinates, post-processing that applies non-maximum suppression to eliminate duplicate detections, and report generation producing a structured analysis summary including detected regions, classification labels, and confidence scores. The pipeline supports multiple model architectures for different imaging modalities, with model selection handled automatically based on image metadata.

8. Database Design

The MAPS database is designed using a normalized relational schema supporting the platform's role hierarchy, AI features, real-time communication, and audit requirements. The schema comprises fifteen interconnected tables covering user management, clinical data, AI predictions, risk assessments, appointments, prescriptions, chat, feedback, and audit logging. Entity Framework Core serves as the ORM layer, providing type-safe data access, automatic migration management, and LINQ-based query capabilities.

Table	Purpose	Key Relationships
Users	All user accounts with role assignments	DoctorProfiles, PatientProfiles
DoctorProfiles	Doctor-specific data (specialization, license)	Users, Assignments, Prescriptions
PatientProfiles	Patient-specific data (blood group, DOB)	Users, Assignments, HealthRecords
Assignments	Doctor-Patient relationship mapping	Doctors, Patients, AuditLog
Appointments	Scheduling data with priority and status	Doctors, Patients, RiskAssessments

HealthRecords	Longitudinal clinical data entries	Patients, Doctors, ClinicalNotes
AIPredictions	Prediction results with confidence scores	Patients, Doctors, DifferentialDiagnoses
DifferentialDiagnoses	Multi-condition ranked diagnosis results	AIPredictions, SuggestedTests
RiskAssessments	Patient risk scores with trend tracking	Patients, Doctors, Appointments
ClinicalNotes	Free-text notes with NLP-extracted entities	HealthRecords, ExtractedEntities
ExtractedEntities	Structured medical entities from NLP	ClinicalNotes, HealthRecords
MedicalImages	Uploaded images with AI analysis results	Patients, Doctors, HealthRecords
Prescriptions	Medication orders with dosage and duration	Doctors, Patients, HealthRecords
ChatMessages	Real-time message persistence	Users (Sender/Receiver)
Feedback	Patient ratings and comments	Patients, Doctors
AuditLog	Tamper-proof action logging	Users (all actions)
Announcements	System-wide and targeted announcements	Admin (creator)

Table 5: MAPS Database Schema - Table Definitions

8.1 Entity Relationships

The database follows a hub-and-spoke pattern centered on the Users table, which connects to role-specific profile tables (DoctorProfiles, PatientProfiles) through one-to-one relationships. The Assignments table serves as the bridge between doctors and patients, implementing a many-to-many relationship with additional attributes such as assignment date, status, and the administrator who created the assignment. Clinical data flows through the HealthRecords table, which links to ClinicalNotes, AIPredictions, MedicalImages, and Prescriptions, creating a comprehensive patient record integrating both human-generated and AI-generated clinical data.

The AI-specific tables follow a hierarchical pattern: AIPredictions links to DifferentialDiagnoses (one prediction can have multiple differential diagnoses), RiskAssessments tracks risk scores over time with timestamp and contributing factor metadata, and ClinicalNotes connects to ExtractedEntities through a one-to-many relationship where each note can produce multiple extracted medical entities. The AuditLog table references Users but is architecturally isolated from clinical data flow, ensuring that audit logging does not interfere with clinical operations and that audit records cannot be accidentally modified through clinical data operations.

9. Security Architecture

The MAPS platform implements a multi-layered security architecture that protects patient data, prevents unauthorized access, and ensures regulatory compliance. Security measures span authentication, authorization, data encryption, audit logging, and network-level protections, creating a defense-in-depth strategy that addresses threats at every level of the technology stack.

9.1 Authentication and Authorization

Authentication is implemented through ASP.NET Core Identity with JWT bearer authentication. Upon successful login, the system issues a short-lived access token (15-30 minutes) and a longer-lived refresh token (7-30 days). The access token contains claims encoding the user's role, assigned identifiers, and permission scope, enabling the authorization middleware to make real-time access control decisions without database lookups. Passwords are stored using bcrypt hashing with automatic salting, and the system enforces configurable password complexity requirements. The Registration Lock mechanism requires administrator approval before newly registered accounts become active, preventing unauthorized self-registration.

9.2 Data Encryption

All sensitive data is encrypted both at rest and in transit. TLS 1.3 encrypts all HTTP communications including API requests, WebSocket messages, and file uploads. Data at rest encryption uses SQL Server Transparent Data Encryption (TDE) for the database and AES-256 encryption for stored files. Personally identifiable information fields in the database can be additionally encrypted at the column level using Entity Framework Core's value conversion features, providing protection even if the database is directly compromised.

9.3 Network and Application Security

The platform implements rate limiting on all API endpoints to prevent brute-force attacks and denial-of-service attempts. CORS policies restrict API access to authorized frontend domains, and Content Security Policy headers prevent cross-site scripting and code injection attacks. Input validation and sanitization are enforced at both the API boundary and the database layer using data annotations, FluentValidation rules, and parameterized queries. Session management policies control maximum concurrent sessions per user and enforce idle timeouts.

10. API Design

The MAPS platform exposes a RESTful API layer built on ASP.NET Core Web API that serves as the communication backbone between the frontend and backend services. All API endpoints follow consistent naming conventions, use standard HTTP methods for CRUD operations, and return

standardized response formats including status codes, error messages, and pagination metadata. API versioning is implemented through URL path versioning to ensure backward compatibility.

API Group	Endpoints	Authorization
Authentication	POST /api/auth/login, /register, /refresh, /logout	Public / Admin-Approved
User Management	GET/POST/PUT/DELETE /api/users, /api/doctors, /api/patients	Admin Only
Assignments	GET/POST/PUT/DELETE /api/assignments	Admin Only
AI Predictions	POST /api/predictions, GET /api/predictions/{id}	Doctor / Admin
Differential Dx	POST /api/predictions/differential, GET /api/predictions/{id}/differential	Doctor / Admin
Risk Assessment	GET /api/risks/patient/{id}, GET /api/risks/alerts	Doctor / Admin
Clinical Notes	POST/GET /api/notes, GET /api/notes/{id}/entities	Doctor / Admin
Medical Images	POST /api/images/analyze, GET /api/images/{id}	Doctor / Admin
Appointments	GET/POST/PUT/DELETE /api/appointments	Doctor / Patient / Admin
Prescriptions	GET/POST/PUT /api/prescriptions	Doctor / Admin
Chat	SignalR Hub /api/chat, GET /api/chat/history/{userId}	Doctor / Patient / Admin
Feedback	POST /api/feedback, GET /api/feedback	Patient / Admin
Audit Log	GET /api/audit, GET /api/audit/{userId}	Admin Only
Analytics	GET /api/analytics/dashboard, /predictions, /users	Admin Only
Announcements	GET/POST/PUT/DELETE /api/announcements	Admin (Write) / All (Read)

Table 6: MAPS API Endpoint Catalog

11. Conclusion

The MAPS platform represents a comprehensive approach to AI-augmented healthcare that prioritizes clinical safety, professional oversight, and patient-centered care. By implementing a strict role hierarchy where Administrators govern the platform, Doctors leverage advanced AI tools for clinical decision support, and Patients access health information through a mediated, secure portal, MAPS ensures that artificial intelligence serves as a powerful enhancement to medical expertise rather than an uncontrolled replacement for it. The five AI-powered Doctor Module features create a unified clinical workflow

where each feature builds upon and enhances the others, from initial symptom analysis through differential diagnosis, risk assessment, appointment prioritization, imaging, and clinical documentation.

The removal of the AI chatbot from the Patient Module reflects a deliberate design philosophy: AI-generated medical insights must always be contextualized and validated by qualified professionals before reaching patients. This approach mitigates the risks of misinterpretation, self-diagnosis, and medical liability while ensuring that patients still benefit from AI-augmented care through their assigned doctors. The platform's modular architecture supports future expansion including wearable device integration for continuous vital sign monitoring, telemedicine video consultations, predictive population health analytics, and integration with national electronic health record systems.